

Regime-Dependent Price Impact in the MOEX Limit Order Book

Nikolay Tagin

May 2026

Abstract

This paper studies whether short-horizon price impact in the Moscow Exchange limit order book depends on the liquidity state of the market. This question matters because average intraday regressions can hide short periods when the book becomes expensive and fragile. I reconstruct top-of-book quotes and order-flow imbalance (OFI) from MOEX Full Orders Log data for 15 blue-chip equities from April to June 2025, then estimate a three-state hidden Markov model on 30-second stock-day sequences. The emissions equation combines de-seasonalized relative spreads and winsorized midprice returns, with a state-specific slope on depth-scaled OFI. The model identifies Calm, Normal and Stressed liquidity-cost regimes. The Stressed regime has the widest spread and thinnest displayed book, but it is not the highest-volatility state. It is more naturally read as liquidity withdrawal than active panic. The price-impact point estimates differ by almost two orders of magnitude between Calm and Stressed. A covariate-transition version of the model shows that lower depth predicts entry into the Stressed state, while higher depth predicts recovery.

Replication repository:

github.com/nikolajtagin/regime-dependent-price-impact-in-the-moex-limit-order-book

The public repository contains the numbered R scripts and generated results used for the tables and figures.

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1 Introduction

The main result is that intraday price impact varies strongly across liquidity states. In the MOEX limit order book, the point estimates imply that a one-standard-deviation shock to depth-scaled order-flow imbalance moves prices far more in the Calm state than in the Stressed state. A single average price-impact coefficient would miss this difference.

This is why a regime model is useful here. Liquidity changes minute by minute as quoted spreads, visible depth and order flow move together. A non-homogeneous HMM lets the model learn hidden states from spreads and returns and then links state transitions to current order-book conditions. In practical terms, it asks what liquidity state the book is in now and which observed conditions make the next state more likely.

I bring these ideas to an intraday equity setting. The empirical sample covers 15 MOEX Blue Chip stocks from April 1 to June 30, 2025. I estimate a non-homogeneous hidden Markov model (NHMM) on 955,110 observations, organized as 930 stock-day sequences on a 30-second grid. “Non-homogeneous” means that transition probabilities are allowed to change with market conditions instead of being fixed for the whole sample. The emissions are bivariate: de-seasonalized relative spread and winsorized midprice return are modeled jointly, with a state-specific return slope on depth-scaled OFI. Transition probabilities depend on de-seasonalized depth, realized volatility (RV), absolute OFI pressure and open/close indicators.

The paper makes three empirical points. The OFI-return slope is strongly state-dependent: the point estimates are 43.5 basis points in Calm, 5.8 in Normal and close to zero in Stressed for a one-standard-deviation OFI shock. Stressed is not the high-volatility regime: quotes widen and depth thins, but realized volatility is closer to Calm and the OFI-return slope is near zero. Depth has the largest and most precisely estimated marginal effect in the transition model: low depth predicts entry into Stressed, while higher depth predicts recovery.

I use the HMM for economic description rather than forecast benchmarking. States are not labelled before estimation. I estimate them and then assign labels by ranking the estimated spread intercepts. The transition covariates are also interpreted carefully. Depth, realized volatility and order-flow pressure are market outcomes as well as predictors, so the coefficients are predictive associations rather than causal effects.

The remainder of the paper covers the literature, data and variables, the HMM specification, diagnostics, results, robustness and conclusion.

2 Literature and Economic Mechanism

2.1 Liquidity, order flow and price impact

Market liquidity has several dimensions. The spread measures tightness, depth measures visible quantity and price impact measures resiliency. Classic adverse-selection models explain why spreads widen when liquidity providers fear informed trading [7, 10]. Kyle [13] gives the standard price-impact interpretation: a higher λ means that a given signed order-flow shock moves prices more strongly. Empirically, Hasbrouck [12] shows that trades contain information about later price movements and Lee and Ready [14] provides a widely used approach for inferring trade direction in intraday data.

The order-book reference most directly used in this paper is Cont et al. [5]. They show that short-horizon price changes are mainly driven by order-flow imbalance at the best bid and ask and that the slope is approximately inversely proportional to market depth. This is why the emission equation uses depth-scaled OFI rather than signed trade volume alone. The estimated λ_i should therefore be read as a local price-sensitivity measure inside a 30-second bucket. It is not a full theory of large institutional execution.

2.2 Dynamic liquidity and fragility

Liquidity is persistent and state-dependent. Andersen and Bollerslev [1] document strong intraday periodicity in volatility. Chordia et al. [4] show commonality in liquidity, which supports looking for market-wide liquidity states rather than only isolated stock-level episodes. Nguyen et al. [18] model intraday liquidity, volume and volatility jointly in the U.S. Treasury market. Their paper is a useful benchmark because it treats liquidity and volatility as connected intraday objects rather than separate daily summaries.

The closest motivation is liquidity fragility. Meldrum and Sokolinskiy [15] study Treasury markets and show that lower market depth increases the fragility of liquidity. The intuition is direct: when few resting orders are visible, the market depends more on quick replenishment. If that replenishment slows down, liquidity can deteriorate even before a large price move is observed. I apply a related hidden-state idea to intraday equity order books and add a regime-specific OFI-return slope, so the model links quoted costs, displayed depth and short-horizon price sensitivity in the same framework.

The liquidity-withdrawal interpretation also connects to theory and evidence on adverse selection and liquidity provision. Foucault et al. [9] model the limit order book as a market where patient traders provide liquidity and impatient traders demand it. Brunnermeier and Pedersen [2] show how market liquidity and funding liquidity can reinforce each other, generating liquidity spirals. Cespa and Foucault [3] show how illiquidity can propagate through cross-asset learning. Raman et al. [19] document that algorithmic liquidity providers can reduce liquidity supply in stressful periods.

2.3 Latent regimes

Hidden Markov models are useful when observed variables are driven by a small number of unobserved states. Hamilton [11] is the canonical reference for Markov-switching methods in economics. In liquidity applications, Flood et al. [8] use hidden Markov chains to identify common liquidity regimes across markets.

I use a custom HMM implementation because the emission equation combines a bivariate state-dependent distribution with a state-specific price-impact slope. I then use an NHMM because a constant-transition HMM can say that regimes exist, but it cannot say which current market conditions make entry into or exit from Stressed more likely. The goal is a compact economic description of intraday liquidity-cost regimes and the conditions associated with transitions between them.

3 Data and Variable Construction

3.1 Sample

The empirical sample is the MOEX Blue Chip equity universe over April 1–June 30, 2025. MOEX describes the Blue Chip Index as a price-return index composed of 15 liquid Russian equity-market stocks, with constituents revised quarterly [16]. The sample contains the following securities:

CHMF	GAZP	GMKN	HEAD	LKOH
MOEX	NLMK	NVTK	PLZL	ROSN
SBER	SNGS	T	TATN	YDEX

The main continuous session is 10:00:00–18:39:59 Moscow time. Weekends are excluded. The primary HMM sample contains 62 calendar trading days, 15 stocks, 930 stock-day sequences and 955,110 final 30-second observations.

MOEX has a brief intraday clearing session at 14:00–14:05 Moscow time. The resulting spread spike and dip in trade activity are removed from the modeled spread and depth levels by within-stock five-minute time-of-day demeaning, but the pattern remains visible in the raw intraday profiles reported in the appendix.

The empirical pipeline is fully reproducible from the numbered R scripts in the [public replication repository](#). Script 01 stores shared settings and functions. Script 02 reconstructs the top of the book from event-level MOEX order logs. Script 03 combines stock-day files into the primary 30-second panel and robustness samples. Script 04 produces diagnostics on seasonality, stationarity, normality, regressors and price impact. Script 05 estimates the one-state benchmark, the constant-transition HMM, the NHMM, compact posterior diagnostics and bootstrap average marginal effects. Script 06 estimates the robustness models.

Table 1: Sample construction

Object	Value
Stocks	15
Calendar trading days	62
Stock-day sequences	930
Primary frequency	30 seconds
Final HMM observations	955,110
Observations per stock	63,674
Main session	10:00–18:39:59
Raw reconstruction grid	5 seconds
Response variable	Winsorized midprice return
Flow variable	Depth-scaled OFI

Notes: The table is based on the model-ready 30-second panel generated by the data-construction scripts. Each stock-day is treated as a separate HMM sequence, so overnight gaps and cross-stock boundaries are not modeled as ordinary 30-second transitions.

3.2 Order-book reconstruction

The raw data are MOEX Full Orders Log files. The file layout includes order number, side, event action, price, volume, trade number and trade price fields [17]. The reconstruction code streams the order log security by security and keeps an active visible book indexed by order number. A placement adds visible limit order volume to the book. A cancellation removes the remaining visible volume. A trade reduces the volume of the resting order that was executed. After each event, the best bid and ask are read from the active price-level maps.

Let b_t and a_t denote the best bid and ask at the end of bucket t . Let D_t^{bid} and D_t^{ask} be the displayed quantities available at those two best quotes. The midprice, spread and total top-of-book depth are

$$m_t = \frac{a_t + b_t}{2}, \quad s_t = a_t - b_t, \quad D_t = D_t^{bid} + D_t^{ask}.$$

Thus D_t is not total market depth at all price levels. It is only the volume visible at the best bid plus the volume visible at the best ask. Only rows with positive prices and valid uncrossed, unlocked top-of-book quotes enter the sample. Empty 5-second intervals are retained using the latest reconstructed quote state and zero within-bucket event counts. This matters because a no-trade interval is still an observation of the order book: liquidity may be available even when nobody trades.

Order-flow imbalance is computed from top-of-book changes following Cont et al. [5]. For event n , with best bid b_n , best ask a_n , bid depth D_n^b and ask depth D_n^a , the contribution is

$$e_n = \mathbf{1}_{\{b_n \geq b_{n-1}\}} D_n^b - \mathbf{1}_{\{b_n \leq b_{n-1}\}} D_{n-1}^b - \mathbf{1}_{\{a_n \leq a_{n-1}\}} D_n^a + \mathbf{1}_{\{a_n \geq a_{n-1}\}} D_{n-1}^a.$$

Here $\mathbf{1}_{\{\cdot\}}$ is an indicator, equal to one if the condition inside braces is true and zero otherwise. The formula sums signed depth changes at the best bid and ask. Bid-side depth contributes positively when the best bid weakly improves, because more visible buying interest has appeared near the top of the book. Ask-side depth contributes negatively when the best ask weakly improves, because more visible selling interest has appeared near the top of the book. Positive e_n therefore means net buy-side pressure, while negative e_n means net sell-side pressure.

The bucket-level OFI is $\text{OFI}_t = \sum_{n \in t} e_n$. The primary flow variable divides OFI by displayed top-of-book depth:

$$x_t = \frac{\text{OFI}_t}{D_t},$$

and the HMM internally scales it by its full-sample standard deviation, $\bar{x}_t = x_t / \hat{\sigma}_x$. This depth scaling means that the same raw imbalance is treated as more important when the visible book is thin. After standardization, the reported λ_i coefficients are measured in basis points per one standard deviation of depth-scaled OFI.

Signed trade volume is also constructed as a robustness variable. It is buy-initiated trade volume minus sell-initiated trade volume inside a bucket. The baseline sign follows the idea of Lee and Ready [14]: a trade above the prevailing midpoint is treated as buyer-initiated, a trade below the midpoint is treated as seller-initiated and trades at the midpoint use the tick rule based on the previous trade price. The MOEX BUYSELL field is saved for checks, but it is not used as the main aggressor-side flag because the same trade number, `TRADENO`, often appears in more than one raw row. To avoid double-counting the same trade, the pipeline first keeps one contribution per `TRADENO` and only then adds trade volume to the bucket.

3.3 Thirty-second model variables

The primary HMM is estimated at 30-second frequency. This choice is a compromise. A 5-second grid has more observations, but many buckets have exactly zero return and prices move in visible tick steps. That is difficult for a Gaussian emission model. A 60-second grid smooths microstructure noise, but it can blur regimes whose expected dwell times are only one to two minutes. The 30-second grid gives a small amount of aggregation before the HMM applies its conditional Gaussian approximation. It does not make returns literally normal: in the primary sample the median return is still exactly zero, so the zero-return mass remains an important diagnostic issue. The 5-second trade-bucket and 60-second models are therefore used as robustness checks.

The midprice return is measured in basis points:

$$r_t = 10^4 (\log m_t - \log m_{t-1}).$$

Returns are winsorized within stock at the 0.5% and 99.5% percentiles, producing r_t^w . This keeps occasional extreme observations from dominating a Gaussian emission model.

The relative spread is $rs_t = s_t/m_t$. The transformed spread removes each stock's own intraday pattern:

$$\tilde{s}_t = \log(rs_t) - \hat{\mu}_{s,c,\tau(t)},$$

where c is the stock and $\tau(t)$ is the five-minute time-of-day bin. Unlike realized volatility and absolute OFI, $\tilde{s}_t$ is demeaned but not divided by its within-bin standard deviation. This preserves economically meaningful differences in spread volatility across stocks while removing deterministic intraday levels.

The remaining transition variables are transformed depth, realized volatility (RV) and absolute OFI pressure. RV means recent return variation. I measure it as the sum of squared returns over the previous 12 buckets:

$$RV_t = \sum_{h=1}^{12} r_{t-h}^2,$$

which is a six-minute window at 30-second frequency. The current return is not included, so RV_t is predetermined relative to the return in bucket t .

The transition covariates are then constructed as

$$\begin{aligned} \tilde{D}_t &= \log(D_t) - \hat{\mu}_{D,c,\tau(t)}, \\ \widetilde{RV}_t &= \frac{\log(RV_t + \epsilon) - \hat{\mu}_{RV,c,\tau(t)}}{\hat{\sigma}_{RV,c,\tau(t)}}, \quad \widetilde{|\text{OFI}|}_t = \frac{|\text{OFI}|_t - \hat{\mu}_{|\text{OFI}|,c,\tau(t)}}{\hat{\sigma}_{|\text{OFI}|,c,\tau(t)}}. \end{aligned}$$

The transition model also includes an opening dummy and a closing dummy. They control for deterministic session-boundary effects that should not be forced into the hidden state alone.

The opening dummy equals one for buckets ending before 10:30 Moscow time, the first 30 minutes of the continuous session. The closing dummy equals one for buckets ending at or after 18:10, the last 30 minutes.

4 Econometric Model

4.1 Notation

Let $Y_t = (\tilde{s}_t, r_t^w)^\top$ be the observed vector in bucket t and let $S_t \in \{1, 2, 3\}$ be the latent liquidity state. The three states are not named before estimation. After estimation, they are ordered by the spread intercept α_i^s and labelled Calm, Normal and Stressed.

Table 2: Main notation

Symbol	Definition
b_t, a_t	Best bid and best ask at the end of bucket t
m_t	Midprice $(a_t + b_t)/2$
rs_t	Relative spread $(a_t - b_t)/m_t$
D_t^b, D_t^a	Displayed depth at the best bid and best ask
D_t	Total best-level depth, $D_t^b + D_t^a$
$\tilde{s}_t$	De-seasonalized log relative spread
r_t^w	Winsorized log midprice return in basis points
OFI_t	Event-level order-flow imbalance accumulated in bucket t
x_t	Depth-scaled OFI, OFI_t/D_t
$\bar{x}_t$	Depth-scaled OFI divided by its full-sample standard deviation
RV_t	Realized volatility, measured as the sum of squared returns over the previous 12 buckets
S_t	Latent liquidity-cost state
λ_i	State-specific price-impact slope on scaled depth-adjusted OFI
Σ_i	State-specific covariance matrix of spread and return residuals
ρ_i	State-specific residual correlation between spread and return equations
$\mathbf{1}_{\{\cdot\}}$	Indicator equal to one when the condition in braces is true and zero otherwise
X_t	Transition covariates $(1, \tilde{D}_t, \widetilde{RV}_t, \widetilde{ \text{OFI} }_t, I_t^{\text{open}}, I_t^{\text{close}})$

4.2 Bivariate emission equation

The HMM observes two variables in each 30-second bucket:

$$Y_t = (\tilde{s}_t, r_t^w)^\top.$$

The first emission is the transformed spread $\tilde{s}_t$, which captures quoted trading cost. The second emission is the winsorized return r_t^w , which captures the short-horizon price move. The model treats these two variables jointly because a liquidity state should describe both the cost of trading and the way prices react to order flow.

Conditional on state i and depth-scaled OFI, the two-dimensional emission is assumed to be Gaussian:

$$Y_t \mid S_t = i, x_t \sim \mathcal{N} \left(\begin{pmatrix} \alpha_i^s \\ \alpha_i^m + \lambda_i \bar{x}_t \end{pmatrix}, \Sigma_i \right),$$

where $\bar{x}_t = x_t/\hat{\sigma}_x$ is the standardized depth-scaled OFI variable used in estimation. The mean vector has two parts. The spread mean is α_i^s , so each state has its own usual spread level. The return mean is $\alpha_i^m + \lambda_i \bar{x}_t$, so each state has its own price-impact slope λ_i . A larger λ_i means that the same standardized OFI shock is associated with a larger return move.

The state-specific covariance matrix is

$$\Sigma_i = \begin{pmatrix} (\sigma_i^s)^2 & \rho_i \sigma_i^s \sigma_i^m \\ \rho_i \sigma_i^s \sigma_i^m & (\sigma_i^m)^2 \end{pmatrix}.$$

The terms σ_i^s and σ_i^m measure how much the spread and return residuals vary inside state i . The parameter ρ_i measures whether unusually high spread residuals tend to arrive together with unusually high return residuals inside the same state. In the estimated model, ρ_i is close to zero in every state (Table 6), so the two residuals are almost independent after conditioning on the state. The joint model is still useful because the state itself is assigned using both pieces of information at once. If the spread and return equations were estimated separately, the paper would lose this single shared state classification.

Equivalently,

$$\tilde{s}_t = \alpha_{S_t}^s + \varepsilon_t^s, \quad r_t^w = \alpha_{S_t}^m + \lambda_{S_t} \bar{x}_t + \varepsilon_t^m.$$

The bivariate structure is kept because spread stress and price-impact stress are not identical. A one-variable model could identify high-spread states or high-volatility states, but it would not directly identify regimes in which quoted costs and sensitivity to order flow change together.

The Gaussian assumption is a working approximation. Raw high-frequency returns have heavy tails, so they are not normally distributed in the unconditional sample. The model only needs a weaker approximation: after 30-second aggregation, winsorization, OFI conditioning and state conditioning, the residuals should be regular enough for the HMM to give a useful classification. Section 5.3 checks this point directly; it is not treated as proof of literal normality.

4.3 Transition equation

The constant-transition HMM is estimated first:

$$p_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i), \quad i, j \in \{1, 2, 3\}.$$

It gives a benchmark for persistence and state interpretation. The expected duration of state i in buckets is $1/(1 - p_{ii})$.

The NHMM then allows transition probabilities to vary with observable conditions:

$$p_{ij,t} = \mathbb{P}(S_{t+1} = j \mid S_t = i, X_t) = \frac{\exp(\beta_{ij}^\top X_t)}{\sum_{\ell=1}^3 \exp(\beta_{i\ell}^\top X_t)}.$$

For each origin state i , the Calm destination is the baseline category, so $\beta_{i1} = 0$. This is only a normalization: multinomial logits need one destination to be the reference category.

The covariate vector is

$$X_t = (1, \tilde{D}_t, \widetilde{RV}_t, \widetilde{|\text{OFI}|}_t, I_t^{\text{open}}, I_t^{\text{close}})^\top.$$

The timing is $X_t \rightarrow S_{t+1}$. Depth, realized volatility, absolute OFI and time-of-day indicators are measured using information available by the close of bucket t .

4.4 Hypotheses

The empirical hypotheses are designed around the central economic question: what mechanism makes transition probabilities depend on observable variables? Let $p_{CS,t}$ denote the probability of moving from Calm to Stressed and let $p_{SC,t}$ denote the probability of moving from Stressed to Calm. In the formulas below, derivatives are shorthand for average marginal effects in the fitted transition model.

The first hypothesis is depth fragility. Lower depth means fewer visible resting orders, so a small liquidity shock should move the market into the Stressed state more easily. Conversely, higher depth should support recovery:

$$H1 : \quad \frac{\partial p_{CS,t}}{\partial \tilde{D}_t} < 0, \quad \frac{\partial p_{SC,t}}{\partial \tilde{D}_t} > 0.$$

The second hypothesis distinguishes active volatility from liquidity withdrawal. If Stressed were a pure high-volatility panic state, higher realized volatility and higher absolute OFI pressure would raise entry into Stressed. If Stressed is instead a withdrawal state, high realized volatility may route observations toward the active Normal regime, while OFI pressure may mainly delay recovery. In words, H2 asks whether Stressed is “expensive and thin” rather than simply “wildly volatile”:

$$\begin{aligned} H2a : \quad & \frac{\partial p_{CS,t}}{\partial \widetilde{RV}_t} \leq 0, \quad \frac{\partial p_{SC,t}}{\partial \widetilde{RV}_t} < 0, \\ H2b : \quad & \frac{\partial p_{CS,t}}{\partial \widetilde{|\text{OFI}|}_t} \approx 0, \quad \frac{\partial p_{SC,t}}{\partial \widetilde{|\text{OFI}|}_t} < 0. \end{aligned}$$

The third hypothesis is intraday variation. Opening and closing periods may have transition hazards that differ from the rest of the session even after intraday profiles are removed. Because deterministic spread and depth patterns have already been de-seasonalized, I expect the boundary dummies mainly to reduce direct entry from Calm to Stressed. The close may also slow recovery if late-day liquidity replenishment weakens:

$$H3 : \quad \Delta^{\text{open}} p_{CS,t} < 0, \quad \Delta^{\text{close}} p_{CS,t} < 0, \quad \Delta^{\text{close}} p_{SC,t} < 0.$$

The fourth hypothesis is model comparison. If observable market conditions help

explain state dynamics, the covariate-transition model should improve information criteria relative to the constant-transition HMM:

$$H4 : \text{BIC}_{\text{NHMM}} < \text{BIC}_{\text{constant HMM}},$$

with the size of the improvement also reported per observation:

$$\Delta\text{BIC}/N.$$

With 955,110 observations, any non-trivial covariate effect can become BIC-positive, so H4 is best read together with the per-observation ΔBIC reported in [Section 6.3](#).

4.5 Estimation and inference

The model is estimated by a custom Baum-Welch expectation-maximization algorithm. It alternates between two simple ideas. In the expectation step, it uses the current parameters to estimate the probability that each observation belongs to each hidden state. In the maximization step, it updates the emission and transition parameters using those probabilities as weights. The emission step updates state-specific bivariate Gaussian regression parameters. The transition step updates multinomial logits using posterior transition weights. To reduce the risk of stopping at a bad local solution, the constant-transition model is estimated from four different starting values. The states are then relabelled after estimation by ascending α_i^s . The four final log-likelihoods differ by less than 0.3 ([Appendix Table 14](#)). Since the fitted log-likelihood is about -3.09 million, this difference is tiny; the baseline solution is therefore not sensitive to the tested EM starting value.

The initial distribution is set to the stationary distribution implied by the transition probabilities. Each stock-day is treated as a fresh sequence, so this is a convenient normalization rather than a claim about the true opening-state distribution. The within-stock time-of-day demeaning of spreads and depth reduces, but does not remove, this simplification.

Inference for transition effects uses a day-cluster bootstrap. Whole calendar days are resampled, preserving all 15 stocks within the resampled day. This keeps same-day market-wide dependence inside each bootstrap draw. The reported confidence intervals for average marginal effects are based on 200 successful bootstrap replications. The method is practical for serial dependence and same-day cross-sectional dependence, but it does not preserve dependence across adjacent trading days.

The bootstrap targets transition average marginal effects, not the emission slopes λ_i . For that reason, the state-specific price-impact estimates are interpreted as point estimates and order-of-magnitude evidence. A full day-cluster bootstrap for λ_i is a natural

extension, but it is not reported here.

The full reproducible specification is in [script 05 of the replication repository](#).

Listing 1: Core R calls behind the primary HMM and NHMM

```
baseline_fit <- fit_constant_hmm(  
  data = dt,  
  k = cfg$n_states,  
  emission_family = "gaussian",  
  covariance_mode = "full"  
)  
  
nhmm_fit <- fit_nhmm(  
  data = dt,  
  k = cfg$n_states,  
  baseline_fit = baseline_fit,  
  design = transition_design,  
  emission_family = "gaussian",  
  covariance_mode = "full"  
)  
  
entry_ame <- compute_transition_ame(  
  nhmm_fit, transition_design,  
  covariate_name = "depth_tilde",  
  origin = 1L,  
  destination = cfg$n_states  
)
```

5 Diagnostics and Assumptions

5.1 Descriptive diagnostics

[Table 3](#) summarizes the main pre-estimation diagnostics. The raw return r_t is very heavy-tailed, with excess kurtosis about 123. Winsorization reduces the excess kurtosis of the return used in the HMM to about 4.0, but it does not make high-frequency returns Gaussian. The state-conditional QQ plot in [Figure 1](#) and the return histogram in [Appendix Figure 7](#) show the residual tail behavior and the zero-return spike that remain after winsorization.

5.2 Stationarity and persistence

The stationarity checks do not indicate a unit-root problem in the diagnostic subsample, but they should not be overread. Augmented Dickey-Fuller (ADF) tests reject a unit-root null for all main variables at the 1% level. Kwiatkowski-Phillips-Schmidt-Shin

Table 3: Selected diagnostics for model variables

Variable	Mean	Std. dev.	Median	Excess kurtosis
Raw return r_t	-0.0117	5.7156	0.0000	122.97
Winsorized return r_t^w	-0.0131	5.1136	0.0000	4.00
Transformed spread $\tilde{s}_t$	0.0000	0.5097	-0.2387	-0.69
Depth-scaled OFI	-0.6621	200.5144	0.0000	20,991.70
De-seasonalized depth $\tilde{D}_t$	0.0000	1.2488	-0.0086	0.74
Standardized realized volatility	0.0000	0.9992	0.0837	33.97

Notes: All statistics are computed on 955,110 30-second observations. Returns are measured in basis points. The high kurtosis of depth-scaled OFI is why the HMM uses OFI as a conditioning variable rather than as a Gaussian emission.

(KPSS) tests fail to reject level stationarity at the 5% level for the same variables. The R package `tseries` does not always print exact p-values at the edges of the test table: 0.01 means “0.01 or smaller” and 0.10 means “0.10 or larger.” For this reason, Table 4 is best read as evidence that the variables look stationary enough for the HMM, not as precise p-value reporting.

The first-lag autocorrelations are near zero for returns and raw depth-scaled OFI, but positive for spreads, depth, absolute OFI and especially realized volatility. The autocorrelation function (ACF) measures how strongly a variable is related to its own lagged values. Realized volatility is highly persistent at the 30-second frequency: its mean ACF is about 0.90 at lag 1 and 0.17 at lag 20. This persistence is one reason to model latent states rather than treat each bucket as independent.

Table 4: Stationarity and persistence checks

Variable	ADF p-value	KPSS p-value	Mean ACF(1)
Winsorized return	0.01	0.10	-0.021
Transformed spread	0.01	0.10	0.259
Depth-scaled OFI	0.01	0.10	0.031
Standardized absolute OFI	0.01	0.097	0.166
De-seasonalized depth	0.01	0.10	0.327
Standardized realized volatility	0.01	0.070	0.899

Notes: The tests are computed on a 50,000-observation diagnostic subsample. The `tseries` package reports 0.01 and 0.10 as edge values, meaning 0.01 or smaller and 0.10 or larger, respectively.

5.3 Normality and price-impact checks

Unconditional normality is rejected by Jarque-Bera tests, almost mechanically with 955,110 observations. This does not invalidate the model, because the Gaussian HMM does not require raw high-frequency returns to be normal before conditioning. The relevant diagnostic is conditional normality: after 30-second aggregation, winsorization, OFI

conditioning and state conditioning, residuals should be regular enough for a Gaussian emission to give a useful regime approximation.

The Gaussian assumption remains strained in two visible ways. First, the winsorized return distribution has a large point mass at zero, shown in [Appendix Figure 7](#). Second, the state-conditional QQ plot shows remaining tail deviations, especially in the Stressed state. After the HMM is estimated, the Viterbi-decoded standardized return residual kurtosis is 4.22 in Calm, 3.56 in Normal and 4.58 in Stressed. These correspond to excess kurtosis of about 1.22, 0.56 and 1.58, respectively.

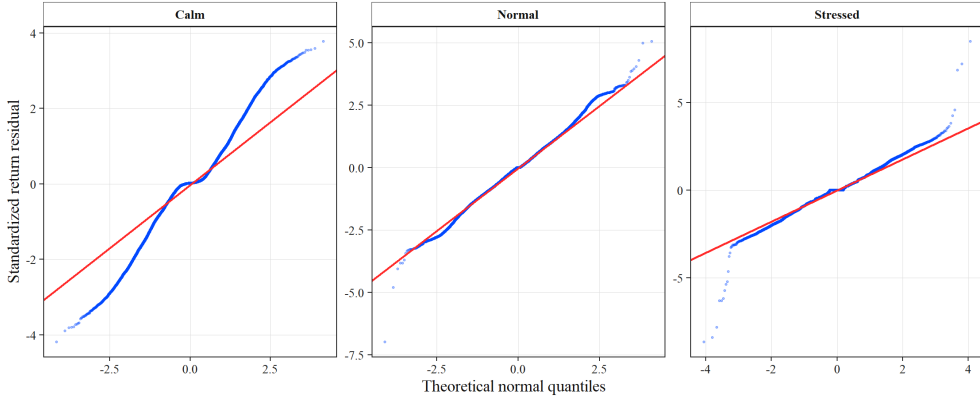


Figure 1: State-conditional QQ plots for standardized return residuals

Notes: The QQ plots are based on standardized residuals from the return emission equation after Viterbi decoding the NHMM state. Normal-state residuals are closest to Gaussian. Calm and Stressed residuals retain visible tail deviations, so the Gaussian model should be read as a transparent classification model rather than a fully specified return distribution.

[Figure 1](#) shows that state conditioning helps, but does not solve all distributional issues. I keep the Gaussian emission because it gives a transparent and stable state structure. The Student- t robustness check in [Section 7.1](#) shows that exact state shares are sensitive to the tail assumption.

A simple pre-HMM price-impact regression also supports the sign convention and the relevance of OFI:

$$r_t^w = \alpha + \lambda x_t + u_t.$$

The slope on depth-scaled OFI is positive and statistically significant under ordinary, heteroskedasticity-robust and day-clustered standard errors ([Table 5](#)). The simple regression explains only a small share of return variation and depth-scaled OFI itself has extremely heavy tails, with kurtosis above 20,000. The HMM therefore uses OFI as a conditioning variable inside state-specific return equations rather than treating it as a Gaussian response.

Table 5: Pre-HMM price-impact regression

Standard error	$\hat{\lambda}$	Std. error	t -stat.	R^2
OLS	0.00296	0.00003	114.34	0.0135
HC1 robust	0.00296	0.00043	6.86	0.0135
Day-clustered HC1	0.00296	0.00045	6.57	0.0135

Notes: The table reports the slope on depth-scaled OFI in $r_t^w = \alpha + \lambda x_t + u_t$, using 955,110 observations. The point estimate is identical across rows, only the standard error and t -statistic change with the assumed dependence structure. The coefficient is in return basis points per one unit of raw depth-scaled OFI, not per one-standard-deviation OFI shock.

6 Empirical Results

6.1 Three liquidity-cost regimes

Table 6 reports the state interpretation. The labels are assigned after estimation by ordering the state spread intercepts from low to high. The Calm state has the lowest spread, highest depth and strongest price-impact slope. The Normal state has moderate spreads and the highest realized volatility. The Stressed state has the widest spread and lowest depth, but low return volatility and a near-zero price-impact slope.

Table 6: Estimated liquidity-cost regimes

State	Share	α^s	Spread bp	Depth	RV	λ	σ_m	ρ
Calm	0.391	-0.415	1.39	5,743	246	43.47	1.93	-0.010
Normal	0.350	0.155	2.76	4,738	669	5.79	7.27	0.003
Stressed	0.260	0.417	3.52	4,381	251	0.12	2.51	0.003

Notes: α^s is the state intercept of the de-seasonalized log relative spread and is used to order states from low to high spread. λ is the coefficient on depth-scaled OFI after scaling by the full-sample standard deviation of that flow variable. It is measured in basis points per one-standard-deviation OFI shock. RV is the state-weighted mean backward-looking six-minute realized variance. State shares are posterior shares from the NHMM.

The key interpretation is that Stressed is a liquidity-withdrawal state. It is stressed in the order book, not mainly in realized volatility. Spreads are wide and depth is low, but price changes are muted and order flow no longer maps strongly into returns. The label should still be read as model-based: Stressed has the heaviest return residual tails, so large individual return shocks do occur inside this state. The regime is more naturally read as liquidity withdrawal than continuous panic selling.

One caution is needed for λ_i , the state-specific price-impact slope. It measures the return response to scaled OFI inside each state, but it also depends on how cleanly scaled OFI works as an explanatory variable in that state. In Calm, depth is high and stable, so depth-scaled OFI is measured more smoothly. In Stressed, depth is lower and changes

more sharply, so the fitted slope can be pushed toward zero even if trading conditions are still poor. This does not overturn the main interpretation, because the same high-spread state also appears in the 5-second trade-bucket, signed-volume and 60-second robustness checks. Still, the Calm-to-Stressed slope gap should be read as a large point-estimate difference, not as an exact structural multiple.

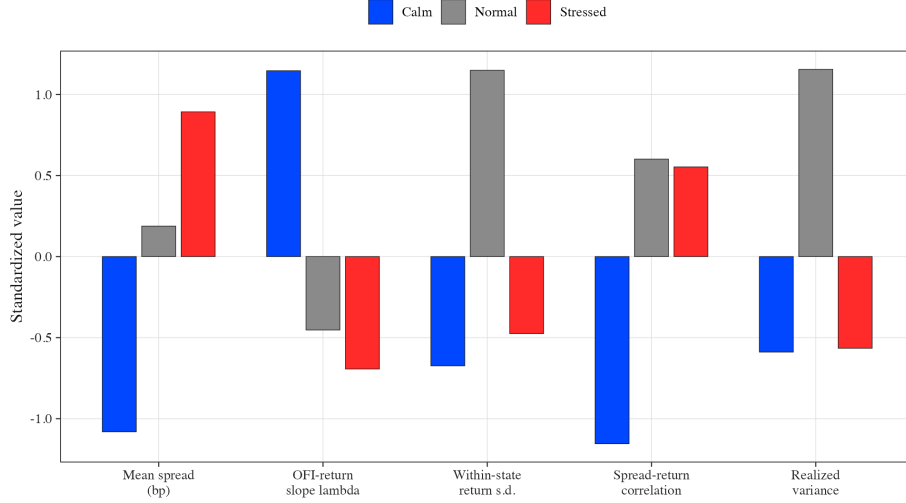


Figure 2: State interpretation: relative position of spreads, price impact, volatility and covariance

Notes: The figure standardizes each metric across states, so the bars show relative rather than raw magnitudes. Calm has the strongest OFI-return slope, Normal has the highest realized and within-state return volatility and Stressed has the widest spread.

6.2 Regime persistence and classification

The constant-transition matrix in Table 7 shows that states are persistent but short-lived. Expected dwell times are 1.40 minutes for Calm, 2.15 minutes for Normal and 1.12 minutes for Stressed. These are minute-scale episodes, not daily or weekly market phases.

Table 7: Constant-transition HMM: transition matrix

Current state	Calm	Normal	Stressed
Calm	0.644	0.148	0.208
Normal	0.138	0.767	0.095
Stressed	0.349	0.096	0.555

Notes: Rows sum to one. The model is estimated on 30-second stock-day sequences.

Classification uncertainty is moderate. A smoothed posterior probability is the model’s probability that a bucket belongs to a given state after using information from the whole stock-day sequence. The maximum smoothed posterior is the model’s confidence in its most likely state for that bucket. The average maximum value is 0.879. About 15.1%

of observations have maximum posterior below 0.70 and 24.6% have maximum posterior below 0.80. Roughly one in four buckets is therefore somewhat ambiguous between states. Decoded paths should be interpreted as useful summaries of posterior probabilities, not as perfectly observed regimes. [Figure 3](#) shows a representative stock-day panel with posterior state probabilities, transformed spread and rolling price impact.

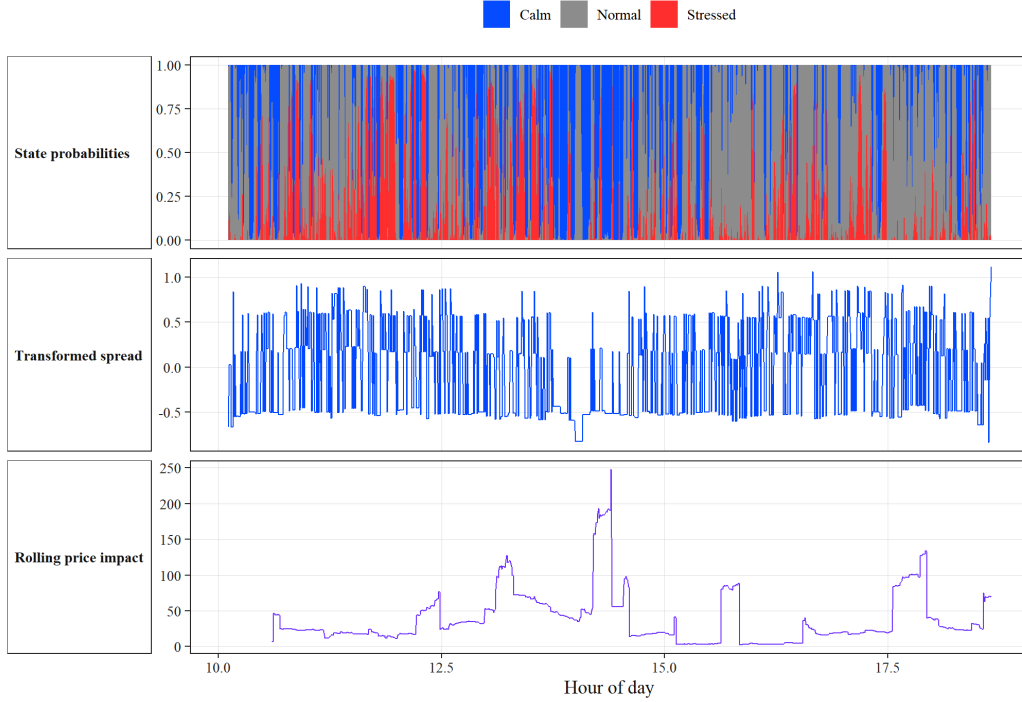


Figure 3: Regime probabilities and liquidity variables on a representative stock-day

Notes: The top panel stacks smoothed probabilities of Calm, Normal and Stressed states. The lower panels show transformed spread and a rolling price-impact estimate. The bottom panel is a right-aligned 60-bucket or 30-minute, no-intercept OLS slope of r_t^w on scaled depth-adjusted OFI, measured in basis points per one-standard-deviation OFI shock. The plot illustrates that state changes are short-lived and often align with changes in quoted liquidity and local price sensitivity.

[Figure 4](#) reports the mean smoothed probability of the Stressed state by time of day. The probability is nearly flat across the day at roughly 25%, with a small opening dip and little end-of-day spike. This differs from the classic U-shape of raw spreads and fits the variable construction: deterministic open/close widening has already been removed from the level variables seen by the HMM.

6.3 Covariate-driven transitions

The covariate-transition HMM improves the fit. [Table 8](#) compares three models on the same 30-second primary sample. Moving from a one-state Gaussian conditional model to a three-state HMM reduces BIC by more than one million points. Allowing transitions to depend on market conditions further reduces BIC by 38,023 points relative

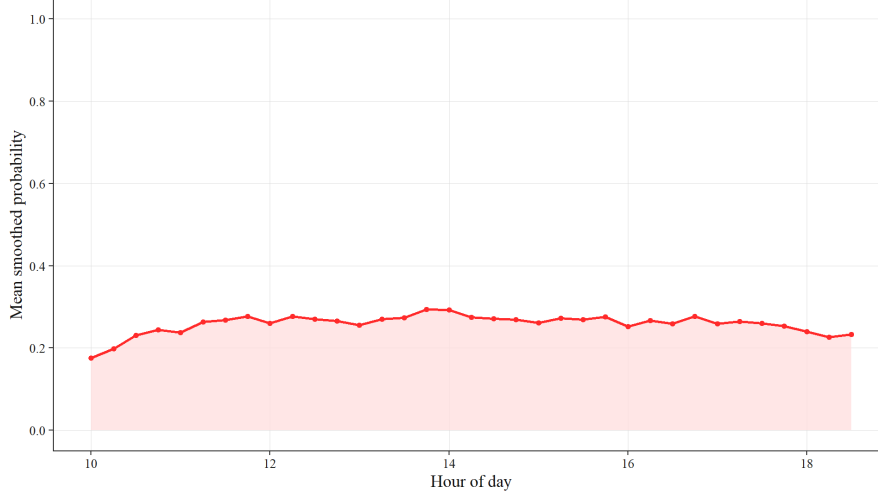


Figure 4: Mean intraday probability of the Stressed liquidity-withdrawal regime

Notes: Smoothed posterior probabilities are averaged over stocks and days in 15-minute bins. The figure is descriptive and uses the completed sample, so it should not be read as a real-time filtered probability.

to the constant-transition HMM. Per observation, this is about 0.04 BIC units, so the improvement is modest but consistent given the large sample.

Table 8: Model comparison on the primary 30-second sample

Model	logLik	Parameters	BIC	Δ BIC
3-state NHMM, covariate transitions	-3,074,937	54	6,150,618	0
3-state HMM, constant transitions	-3,094,155	24	6,188,641	38,023
1-state Gaussian conditional model	-3,618,941	6	7,237,965	1,087,346

Notes: All three models use the same observations and the same response and flow variables. Δ BIC is relative to the best model.

The average transition matrix implied by the NHMM is shown in [Table 9](#). Compared with the constant-transition model, average persistence is lower because the transition probabilities now vary with market conditions. The Stressed state still has a meaningful recovery channel directly to Calm.

Table 9: NHMM average implied transition matrix

Current state	Calm	Normal	Stressed
Calm	0.579	0.216	0.205
Normal	0.181	0.682	0.136
Stressed	0.323	0.153	0.524

Notes: Entries are transition probabilities averaged over the observed covariate distribution.

Average marginal effects are easier to interpret than raw multinomial-logit coefficients. An average marginal effect asks: if one covariate changes while the observed rows

are otherwise kept as they are, how much does the transition probability change on average? Table 10 and Figure 5 show the effect of a one-unit increase in each covariate on two probabilities: entry from Calm to Stressed and exit from Stressed to Calm.

Table 10: Average marginal effects on Stressed-state entry and recovery

Covariate	Entry AME	95% CI	Exit AME	95% CI	Direction
Depth $\tilde{D}_t$	-0.050	[-0.053, -0.048]	0.042	[0.039, 0.046]	H1
Realized volatility	-0.020	[-0.026, -0.014]	-0.053	[-0.062, -0.044]	mixed
Absolute OFI pressure	-0.001	[-0.006, 0.003]	-0.034	[-0.050, -0.022]	H2b
Open dummy	-0.035	[-0.055, -0.017]	-0.002	[-0.024, 0.024]	H3
Close dummy	-0.022	[-0.034, -0.011]	-0.025	[-0.043, -0.006]	H3

Notes: Entry is the change in the probability of moving from Calm to Stressed in the next bucket. Exit is the change in the probability of moving from Stressed to Calm. For continuous variables, the AME is the average effect of a one-unit increase. For the open and close dummies, it is the change from 0 to 1. Confidence intervals are from 200 day-cluster bootstrap replications.

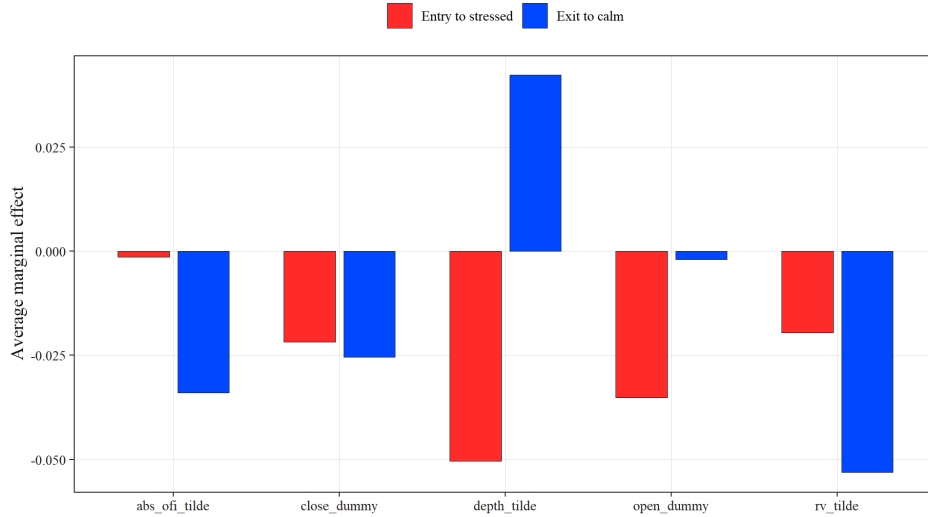


Figure 5: Average marginal effects for entry into and exit from the Stressed state

Notes: Red bars show how each variable changes the chance of moving from Calm to Stressed. Blue bars show how each variable changes the chance of moving from Stressed back to Calm. The most important pattern is depth: thin books predict entry into Stressed, while deeper books predict recovery.

The depth result supports the main fragility mechanism. A one-unit increase in de-seasonalized depth is associated with a 5.0 percentage point lower probability of Calm-to-Stressed entry and a 4.2 percentage point higher probability of Stressed-to-Calm recovery. Among the five covariates, depth has the largest and most stable transition effect in the reported AMEs.

The volatility and OFI effects are more nuanced. Higher realized volatility predicts both lower entry into Stressed from Calm and lower recovery from Stressed to Calm. The first sign supports the idea that Stressed is not mainly a high-volatility entry state. The second is harder to interpret: it may route the market toward Normal or it may make Stressed more persistent once entered. Absolute OFI pressure does not significantly trigger

Stressed entry, but it significantly reduces recovery. This suggests that OFI pressure is more of a propagation mechanism than an ignition mechanism for liquidity withdrawal: it keeps stress from clearing, rather than clearly starting it.

The transition equation should be read as reduced-form. Depth, volatility and OFI pressure are all market outcomes as well as predictors. The estimates therefore describe which observed conditions precede state changes. They do not prove that changing depth alone would causally move the market between regimes.

7 Robustness and Limitations

7.1 Robustness checks

The robustness script estimates five constant-transition HMM variants. These checks target the state model rather than the NHMM transition equation. They ask whether the three-state structure, the OFI flow variable and the 30-second frequency are essential for the main interpretation.

Table 11: Robustness checks

Specification	Converged	State shares	Interpretation
2-state Gaussian, same emissions	Yes	0.379 / 0.621	Two states are too coarse, the three-state structure separates active trading from liquidity withdrawal.
Student- t , $\nu = 5$	No, maxit	0.121 / 0.389 / 0.489	Heavy-tail emissions reshape state shares and need more work, result is a sensitivity check, not a clean confirmation.
5-second trade buckets	Yes	0.410 / 0.420 / 0.170	Three regimes survive at higher frequency, but trade-bucket sampling misses no-trade liquidity-withdrawal periods.
Signed-volume flow instead of OFI	Yes	0.414 / 0.297 / 0.289	The state structure survives, but the fit is worse than the OFI-based primary specification on the same sample.
60-second frequency	Yes	0.406 / 0.305 / 0.289	The regime structure is similar, supporting 30 seconds as a reasonable primary frequency.

Notes: The table asks whether the main state interpretation survives reasonable changes in the model. BIC values are not compared across different sampling frequencies because the observation counts differ. The signed-volume and two-state checks are directly comparable to the primary constant-transition specification because they use the same 30-second sample.

The robustness checks are not uniform, but they support the main three-state interpretation. The three-state structure is identified, OFI is preferred to signed volume in the HMM and the result is not an artifact of choosing exactly 30 seconds. The Student- t check is the main unresolved point. It did not converge in 50 EM iterations and allocates 48.9% of observations to the high-spread state, compared with 26.0% under the Gaussian primary model. This is the largest robustness disagreement. I therefore emphasize the qualitative structure – low-spread Calm, active Normal and high-spread Stressed – more than the exact Stressed-state share.

7.2 Limitations

Several limitations matter. First, the estimates are reduced-form: depth, volatility and order-flow pressure are jointly determined with hidden liquidity states, so the transition effects are predictive associations rather than causal estimates. In other words, the model shows which conditions tend to come before state changes, but it does not prove what would happen if depth could be changed experimentally.

Second, the pooled model constrains emissions and transition dynamics to be common across 15 heterogeneous stocks. The decoded Stressed share varies from about 20% for GAZP to about 41% for HEAD ([Appendix Table 12](#)). Across stocks, those that spend more time in Stressed also tend to have a wider mean spread conditional on being Stressed ([Appendix Figure 6](#)). The pooled HMM cannot exploit this cross-sectional pattern, which is another reason a per-stock or hierarchical specification is the natural next step.

Third, the model uses best-level OFI rather than multi-level order-book imbalance or cross-stock OFI. Multi-level OFI [\[20\]](#) and cross-impact measures [\[6\]](#) would be natural extensions, especially in a pooled blue-chip panel. Finally, trade-sign and OFI reconstruction are necessarily imperfect because exchange order-log conventions require careful interpretation of repeated trade rows and resting-side indicators. The sample covers one three-month period in 2025 and does not explicitly adjust for corporate actions, trading halts or special events beyond the cleaning rules embedded in the reconstructed top-of-book sample.

8 Conclusion

This paper builds an empirical workflow for intraday liquidity-cost states in the MOEX limit order book. The reason for doing this is not only to classify observations. It is to show that the same market can move through different short-lived liquidity-cost regimes and that an average price-impact estimate hides this movement. The code reconstructs best quotes from event-level order logs, builds 30-second stock-day sequences, constructs OFI and de-seasonalized liquidity variables, estimates a three-state HMM and then allows transition probabilities to depend on observable order-book conditions.

The main finding is that price impact is strongly state-dependent. Calm has tight spreads, deep books and a strong OFI-return relation. Normal is the active price-discovery state, with the highest return volatility. Stressed has the widest spreads and thinnest book, but a near-zero price-impact slope and relatively low realized volatility. I interpret it as liquidity withdrawal rather than continuous panic selling.

The transition results support a depth-fragility mechanism. Low depth predicts entry into the Stressed state and higher depth predicts recovery. This is the main value added by the NHMM: it links the hidden liquidity state to observable conditions in the

book instead of only reporting average regime persistence. Realized volatility and absolute OFI pressure mainly affect persistence and routing. The results should be read as reduced-form evidence: they show that observable order-book conditions help identify short-lived liquidity-cost states, but they do not establish a structural causal effect of depth or order flow. The sample period also makes this one of the first intraday liquidity-regime studies of MOEX after the structural change in the Russian equity market that began in 2022, which limits the available comparison set for the depth-fragility result. Future work could estimate stock-specific or hierarchical state dynamics, use multi-level order-book imbalance and revisit the heavy-tail emission with a fully converged Student- t or mixture specification.

Appendix

Table 12: Per-stock Stressed-state shares

Stock	Share	Spread bp (Stressed)	Depth (Stressed)	Avg. max post.
HEAD	0.407	7.67	227	0.900
SNGS	0.347	5.79	34,816	0.873
PLZL	0.338	3.07	306	0.860
TATN	0.307	3.30	786	0.890
YDEX	0.306	3.20	204	0.883
T	0.301	1.61	283	0.894
LKOH	0.280	1.84	170	0.900
NLMK	0.277	3.77	3,647	0.879
CHMF	0.260	4.54	558	0.890
SBER	0.246	0.97	4,339	0.866
ROSN	0.245	2.48	1,150	0.885
GMKN	0.220	3.88	6,056	0.862
MOEX	0.215	1.99	1,537	0.858
NVTK	0.210	3.86	717	0.867
GAZP	0.197	1.86	5,660	0.884

Notes: The share is the fraction of a stock's observations decoded as Stressed. Spread and depth are averaged only over observations decoded as Stressed, so they describe what Stressed looks like for each stock. The λ coefficients are not shown by stock because the primary model estimates one pooled set of state slopes for all 15 stocks.

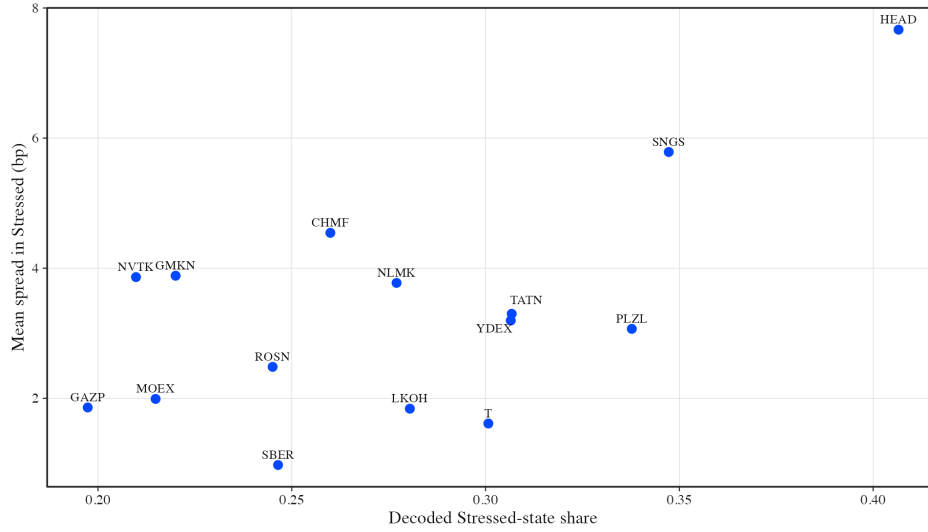


Figure 6: Per-stock Stressed-state share and Stressed-state spread

Notes: Each point is one stock. The horizontal axis shows how often the stock is decoded as Stressed; the vertical axis shows its average spread during those Stressed buckets. The upward pattern means that stocks spending more time in Stressed also tend to have more expensive trading conditions when Stressed.

Table 13: State-conditional residual kurtosis

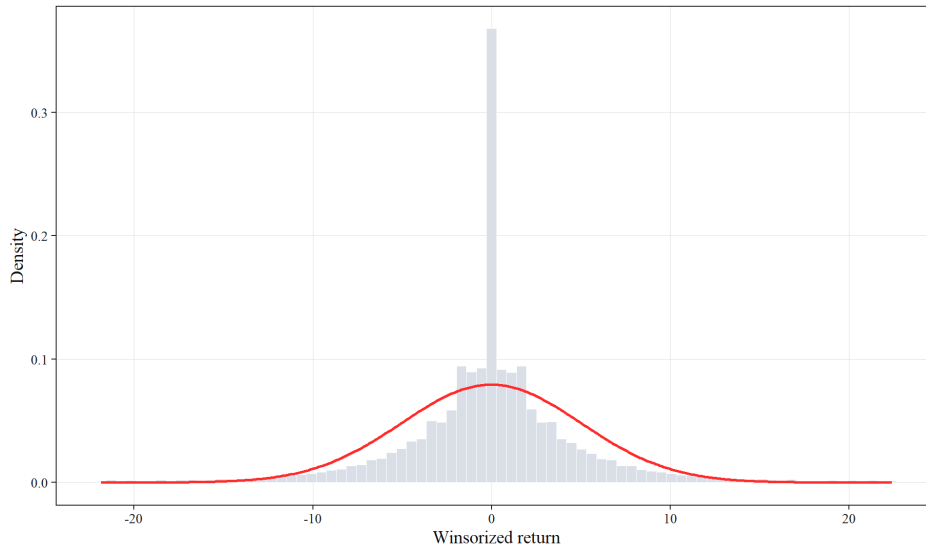
State	Return kurtosis	Spread kurtosis	Observations
Calm	4.22	2.34	390,993
Normal	3.56	2.22	299,454
Stressed	4.58	3.20	264,663

Notes: Kurtosis measures tail thickness. A normal distribution has kurtosis 3, so values above 3 mean heavier tails than normal. The return residuals therefore still have moderate tail excess after state conditioning, especially in Calm and Stressed.

Table 14: Baseline HMM convergence across starting values

Start	Log-likelihood	Iterations	Convergence code
1	-3,094,155.40	71	0
2	-3,094,155.54	71	0
3	-3,094,155.33	71	0
4	-3,094,155.44	71	0

Notes: Each row starts the EM algorithm from a different initial guess. The final log-likelihoods are almost identical, which means the baseline HMM solution is not driven by one lucky starting value. States are relabelled after estimation by ascending spread intercept.

**Figure 7:** Winsorized returns and normal overlay

Notes: The tall central spike means many 30-second buckets still have exactly zero return. This explains why the Gaussian emission is used as a practical classification approximation, not as a literal claim that all intraday returns are normally distributed.

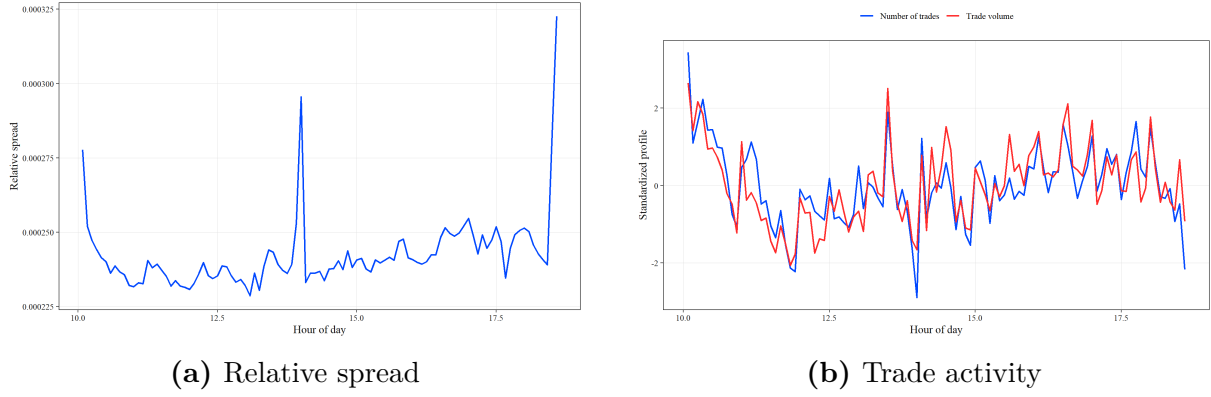


Figure 8: Intraday profiles before state estimation

Notes: The 14:00 movement is caused by the MOEX intraday clearing interval. These raw profiles show why time-of-day adjustment is needed: without it, the HMM could confuse regular daily patterns with genuine liquidity regimes. Times are Moscow time.

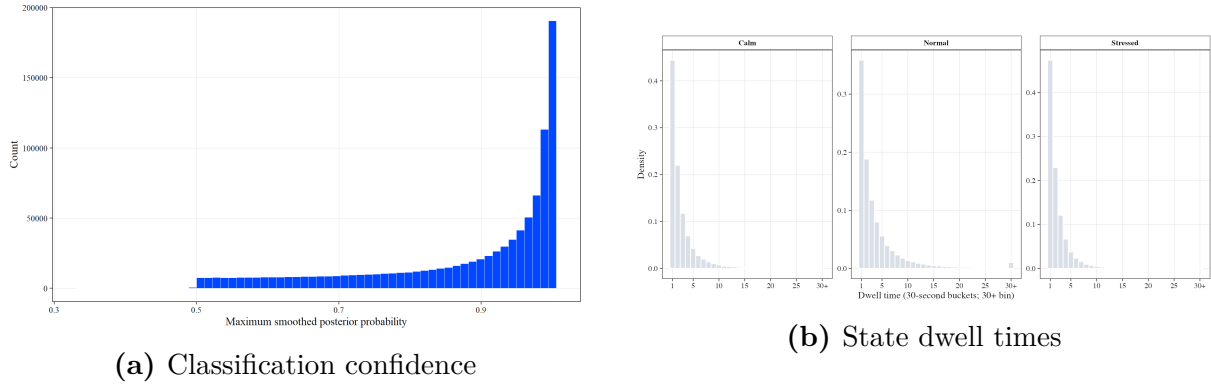


Figure 9: Supplementary classification diagnostics

Notes: Panel (a) shows how confident the model is in its most likely state assignment for each bucket. Panel (b) shows how long decoded state episodes last, with longer episodes grouped at 30 buckets so the common short episodes remain visible. Together, the panels show that the states are useful but not perfectly certain, and that most episodes last minutes rather than hours.



Figure 10: Correlation heatmap for transition regressors

Notes: This heatmap checks whether the transition regressors mostly repeat the same information. The correlations are not large enough to make the covariates redundant, so depth, realized volatility, absolute OFI pressure and open/close indicators are kept as separate transition variables.

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